

Tribhuvan University  
Faculty of Humanities & Social Sciences  
OFFICE OF THE DEAN

2021

**Bachelor in Computer Applications**

Course Title: **Distributed System**

Code No: **CACS352** Semester: **VI**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

*Candidates are required to answer the questions in their own words as far as practicable.*

**Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

**Group B**

2. What is RPC? Explain the RPC process in brief.
3. Explain different characteristics of Distributed System.
4. Why is clock synchronization needed in a Distributed system? Explain the Network Time Protocol in brief.
5. Explain structured naming scheme of an entity in brief.
6. Explain different types of distributed systems.
7. How is process resilience achieved through groups?
8. Write short notes on (Any Two): a) Bully Algorithm b) Berkeley Socket c) Denial of Service

**Group C**

*Attempt any TWO questions[2x10=20]*

9. What is virtualization? Explain different types of virtualization. Explain the role of middleware in a distributed system.
10. What are different architecture styles of distributed systems? Explain each of them in detail.
11. What is the role of a coordinator in a distributed system? Explain different approaches of electing a coordinator.